

Teaching Exam

Subject – Computer Science

EMERGING TRENDS

Lecture No –1

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Topics to be Covered

Topic

Emerging Trends



→ Artificial Intelligence.

→ Big data.

→ Natural Language Processing.

→ IoT.

→ Immersive Experience.

→ Virtual Reality.

→ Augmented Reality.

→ Robotics.

When human intelligence is embedded in machine.
by some technology that is called Artificial Intelligence.

Chat Gpt

- Please make a roadmap to score full marks in Exam.
- Tokens
- Contextual Understanding
- Knowledge Retrieval
- Response Generation
- Rephrasing

Big Data →

Size matters of data

Small Data

< 10GB

Excel; File System (computer)

Medium Data

10GB < — < 1TB

⇒ DBMS
(Hard disk)

Big data

< 1TB

Hadoop, Distributed DBMS

Big Data characteristics $\rightarrow$ 5V.

$\rightarrow$ Volume $\rightarrow$

$\rightarrow$ Velocity $\rightarrow$

$\rightarrow$ Variety $\rightarrow$

$\rightarrow$ Veracity $\rightarrow$ not all data is useful

$\rightarrow$ Value $\rightarrow$

IIoT stands for Internet of Things. Connecting everyday things embedded with electronics, software, and sensors to the internet enables them to collect and exchange data.

Benefits of IoT:

- Efficient resource utilization ✓
- Minimizing Human effort ✓
- Saves time ✓
- Development of AI through IoT ✓
- Improved security ✓

IoT
↳ Internet of Things

IoT Features:

- Connect various things to the IoT Platform
- Analyze the data collected and use it to build business intelligence
- Integrate various models to improve user experience.

IoT Ecosystem:

Basic architecture:

- **Application layer:** Responsible for delivering application-specific services to the user. Defines an application in which the IoT can be deployed.
- **Network layer:** Responsible for connecting smart things, network devices, and servers. Also used for transmitting and processing sensor data.
- **Perception layer:** Sensors sense and gather information about the environment. Senses physical parameters or identifies other objects in the environment.

Application of IoT:

- Energy conservation
- Smart Transport
- Social life and entertainment
- Supply chain and logistics
- Health and fitness
- Smart environment
- Home automation
- Smart agriculture

IoT Sensor:

It is a device that captures real-world data and translates it into a piece of information that could be interpreted by other instruments. As the name suggests, sensors react to stimuli – be it changes in temperature, sound, or different properties. Hence a sensor in an IoT system senses the desired physical quantity and converts it into electrical signals transmitted to the central cloud-based server directly or via an on-site microcontroller.

Types of Sensors:

- Temperature sensor
- Humidity sensor
- Proximity sensor
- Level sensor
- Gyroscope.

1. What is the full form of IoT?

- a) Internet of Technology
- b) Incorporate of Things
- c) Internet of Things
- d) Incorporate of Technology

2. What is IoT?

- a) network of physical objects embedded with sensors
- b) network of virtual objects
- c) network of objects in the ring structure
- d) network of sensors

3. Who coined the term "Internet of Things"?

- a) Kevin Aston ✓
- b) John Wright
- c) Edward Jameson
- d) George Garton



2 mins Summary



TEACHING
WALLAH

Topic

IOT - I

Artificial Intelligence

Big data → 8V.

IOT → shall

THANK YOU